

Pooja Kiran

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EDUCATION

Arizona State University

M.S. in Information Technology (Security), GPA: 3.87/4.00

Tempe, AZ

Aug. 2024 – May 2026

M. S. Ramaiah University of Applied Sciences

B.Tech. in Computer Science & Engineering, CGPA: 8.44/10

Bengaluru, India

Aug. 2019 – Aug. 2023

EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 – Present

Self-Directed Research

Tempe, AZ

- Built security tooling for agent tool-call enforcement, AWS IAM analysis, prompt-injection testing, model supply-chain review, training-data poisoning detection, and adversarial ML.
- Developed Python systems with policy enforcement, SARIF, GitHub Code Scanning, Elastic telemetry, CI security gates, and reproducible adversarial evaluation.
- Verified flagship tooling with **659 MCP tests at 82% coverage**, **235 IAM tests**, and **173 LLM red-team tests at 95.15% coverage**.

Business & Compliance Lead | AEROSSEC Externship

Aug. 2025 – Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business, compliance, and third-party security analysis for AEROSSEC, a proposed protection layer for airline Passenger Service System integrations.
- Built a **\$120K first-year commercialization scenario** and five-year financial model; presented the business and compliance case to ASU and Honeywell stakeholders.

Graduate Teaching Assistant - IT Grader

Jan. 2025 – Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Reviewed approximately **85 web-development submissions per semester** for implementation quality, debugging, code correctness, and secure coding practices.
- Assessed **52 graduate cybersecurity assignments** on security policy, compliance, risk analysis, and information-security controls.

PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC, Elastic Security*

Jul. 2026 – Sep. 2026

- Built an inline MCP/JSON-RPC gateway with **55 prompt-injection patterns**, capability and PII/exfiltration checks, fail-closed policy enforcement, rate limits, and hash-chained audit logs.
- Validated the gateway with **659 passing tests**, **82% statement coverage**, **9 Elastic Security rules**, and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Aug. 2026 – Sep. 2026

- Developed an IAM analyzer with **25 deterministic rules** for wildcard access, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy risks, and audit tampering.
- Added **SARIF 2.1.0**, GitHub Code Scanning, and CI exit gates; validated **235 passing tests**, with **3 live-scan tests skipped** without AWS credentials.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, SARIF*

Jul. 2026 – Sep. 2026

- Implemented a non-executing model supply-chain scanner for pickle, SafeTensors, GGUF, ONNX, Keras, source code, dependencies, provenance, and impersonation signals.
- Detected **33/33 adversarial fixtures** with **0 actionable false positives across 4 benign samples** in the committed regression suite.

LLM Red-Team Framework | *Python, FastAPI, scikit-learn, SARIF*

Aug. 2026 – Sep. 2026

- Designed an offline prompt-injection evaluation framework spanning **6 adversarial categories** with grouped-template and novel-phrasing OOD testing.
- Measured **0.9714 grouped F1 vs. 0.7188 OOD F1**; verified **173 passed**, **1 skipped**, and **95.15% coverage**.

TECHNICAL SKILLS

Security: Agent Security, LLM Security, Prompt Injection, AWS IAM, Application Security, Threat Modeling, Model Supply Chain, Adversarial ML, Elastic Security

Tools: Python, FastAPI, PyTorch, scikit-learn, AWS, MCP/JSON-RPC, SARIF, GitHub Actions, Docker, Kubernetes, CodeQL, Prometheus, MITRE ATT&CK/ATLAS